

Tree growth responses to extreme drought events are not well predicted by climate

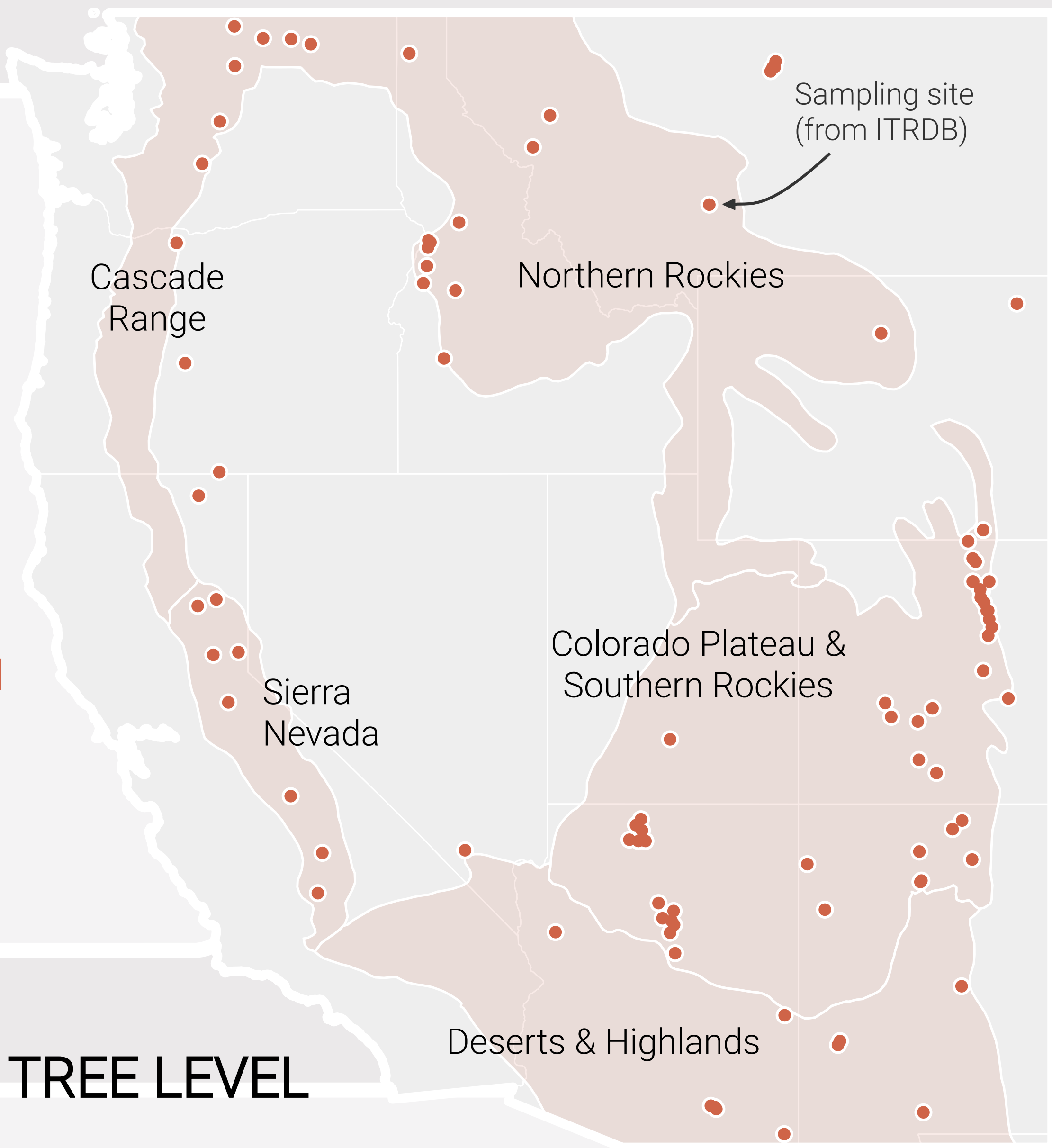
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OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture
- Combined effects of these drivers on tree growth remain poorly understood at large scales
- A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
- Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
- Current models likely **underestimate growth declines** during extreme events

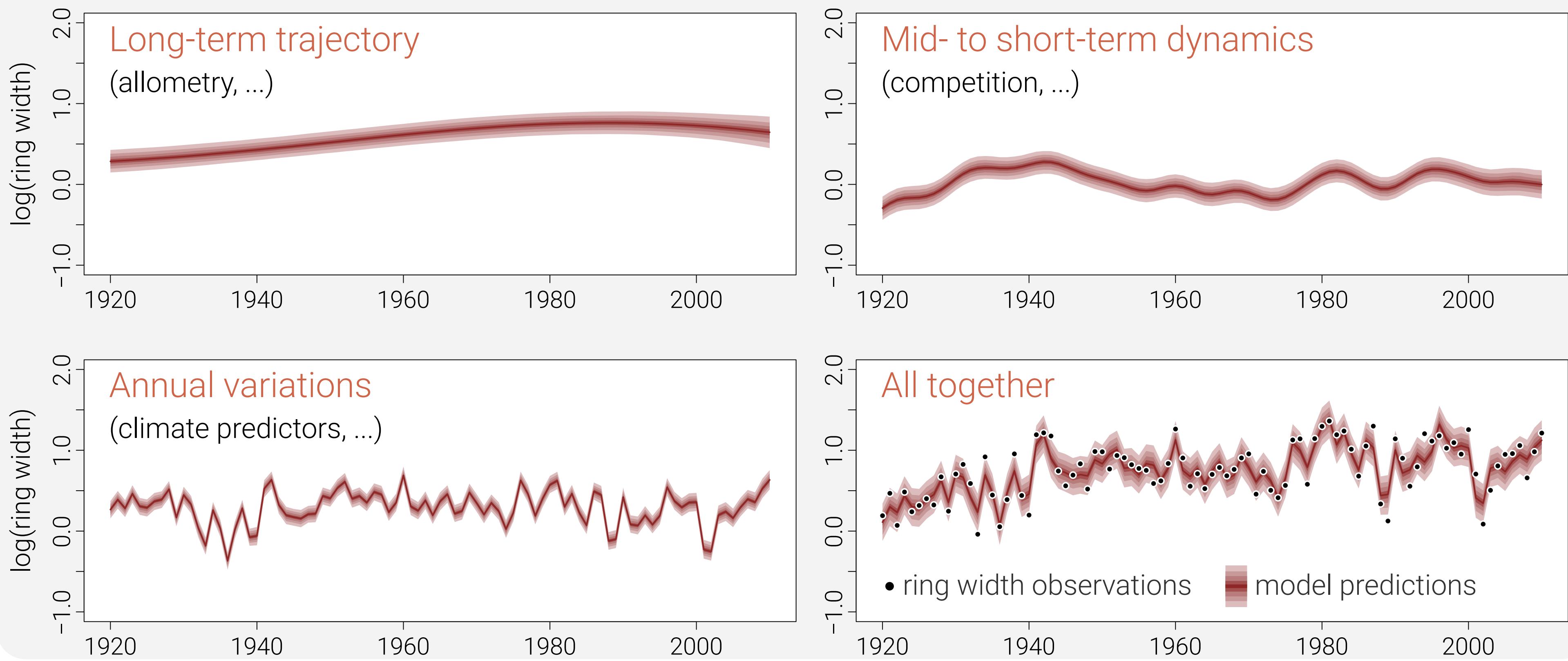
1. BACKGROUND

- Tree-ring width time series are **one of the longest and most spatially extensive** records of forest response to climate
- However, tree rings are affected by **many drivers other than climate**: age-size effects, competition, biotic disturbance...
- Standard dendrochronology approaches detrend individual trees and aggregate them to identify some shared climatic signal
- But **drought stress operates at the individual level**, whose responses integrate non-linear physiological processes
- To capture how extreme events, we need to model individual trees explicitly



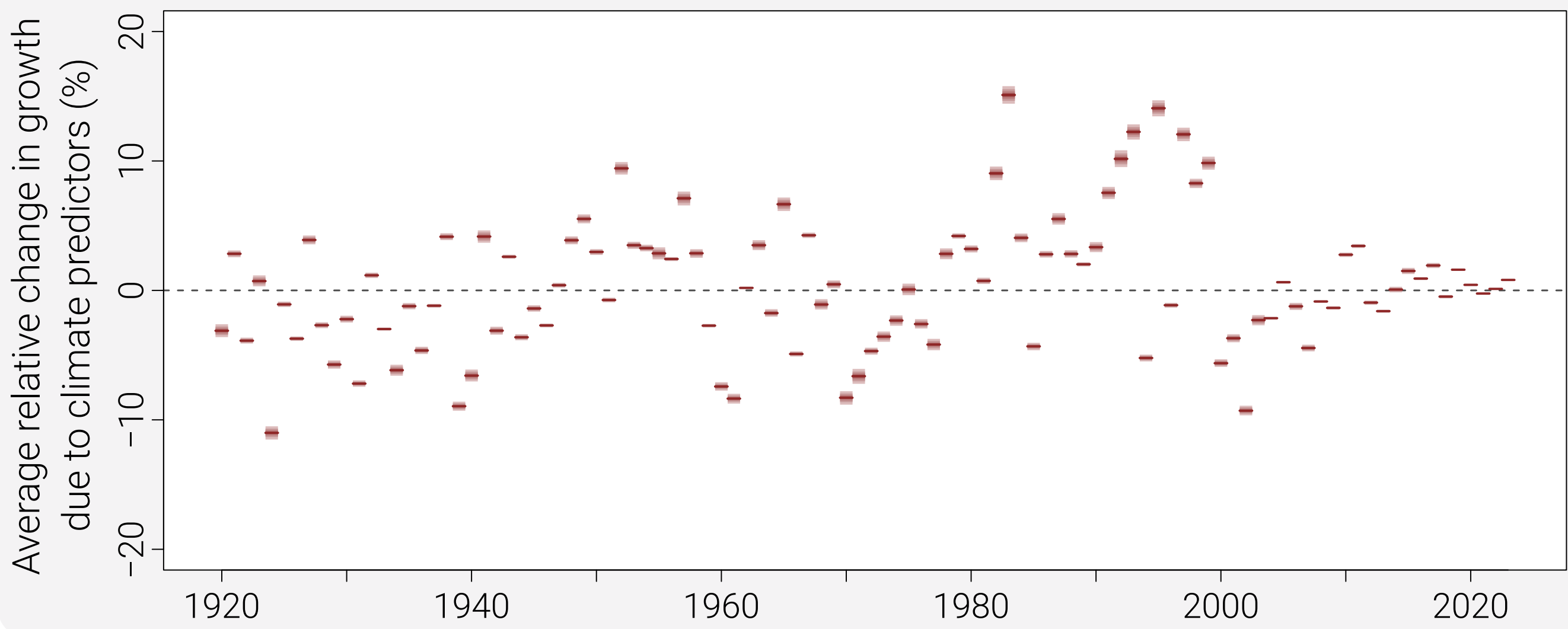
2. MODELING GROWTH AT THE TREE LEVEL

Tree growth results from three major components:



3. MAIN CLIMATIC DRIVERS OF GROWTH

Growing degree days, winter precipitation, and vapor pressure deficit together explain **±10% of annual variation in growth**

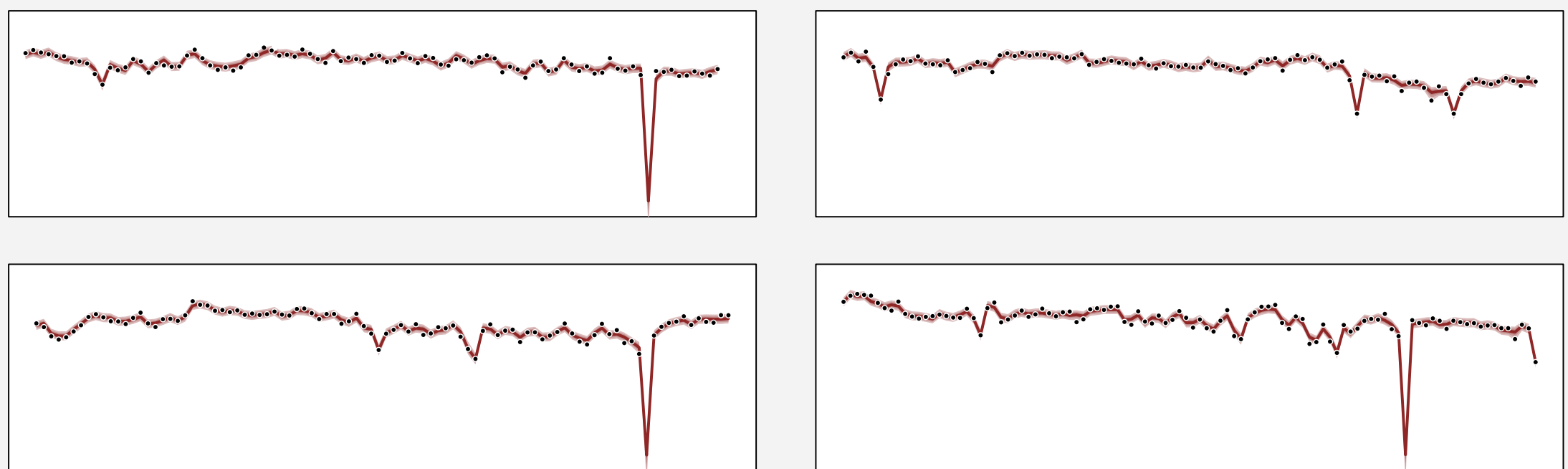
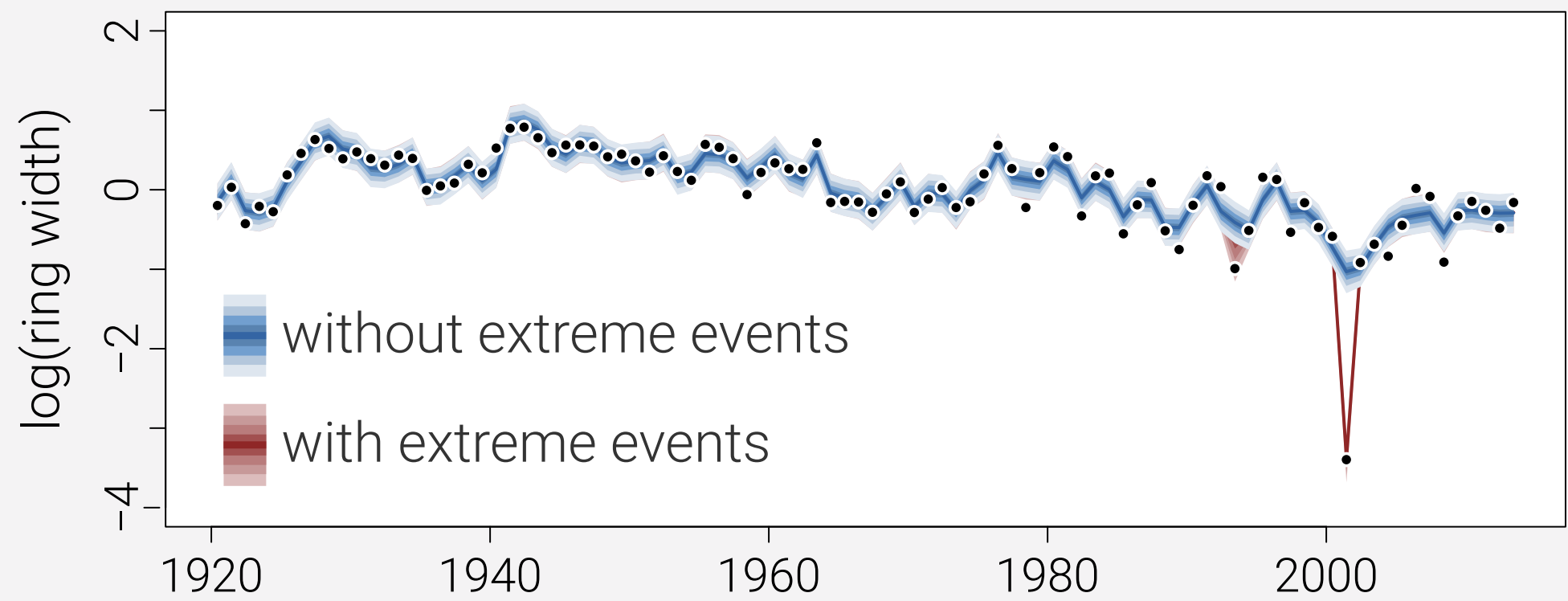


Our climate-growth component generally captures some known **wet periods** (1980-1990s) and some **less favorable periods** for growth (e.g. 1930s, post-2000).

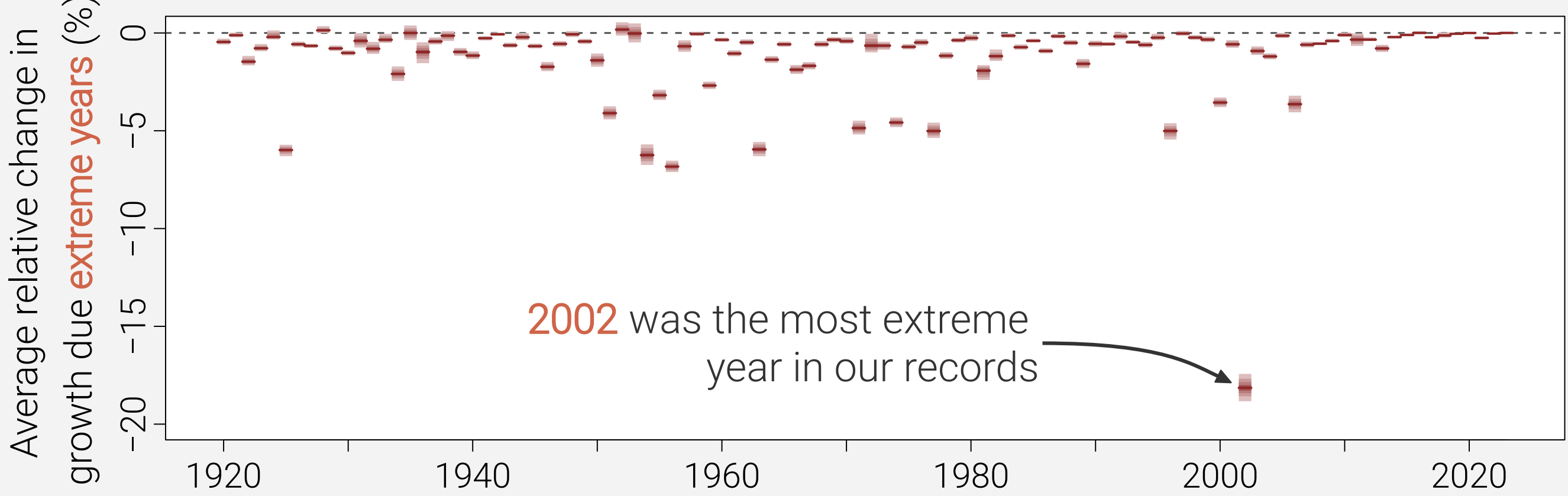
4. EXTREME GROWTH REDUCTIONS (SHOCKS)

A tree can show **severe growth suppression**, far beyond what its climate response predicts.

When one tree experiences a shock, **neighboring trees within the same forest** tend to as well.



These shocks have major **consequences at the ecosystem scale**.



The synchrony of shocks reveals a shared, forest-level signal, consistent with a **common external stressor** (rather than individual chance).

Many of these shocks correspond to known major drought events, which **suppress growth by up to almost 20%** on top of the effects of climate predictors.

5. IMPLICATIONS FOR FORESTS AND FORECASTS

Classical dendroclimatology extracts climate signals by averaging out individual variation (top-down), while here **we model individual tree growth** (bottom-up). Climatic drivers capture inter-annual to decadal variability, but **systematically miss extreme growth shocks** that are due to individual endogenous processes. Accounting for these **non-linear, individual-level responses is critical for accurate forecasts** of forest vulnerability and carbon storage under climate change.

References...